

Task 2: End-to-End ML Pipeline with Scikit-learn Pipeline API

Objective:

Build a reusable and production-ready machine learning pipeline for predicting customer churn.

Dataset:

Telco Churn Dataset

Instructions:

- Implement data preprocessing steps (e.g., scaling, encoding) using Pipeline
- Train models like Logistic Regression and Random Forest
- Use GridSearchCV for hyperparameter tuning
- Export the complete pipeline using joblib

Skills Gained:

- ML pipeline construction
- Hyperparameter tuning with GridSearch
- Model export and reusability
- Production-readiness practices